QUESTION BANK
PCCST501 – COMPUTER NETWORKS
MODULE 2

Transport Layer • Network Layer

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1. Syllabus and examination alignment

Module 2 carries **18 contact hours**. It covers the Transport Layer (services, protocols, UDP and TCP) and Network Layer (introduction, network-layer protocols, unicast routing, multicast routing, next-generation IP and Quality of Service). The hands-on component includes sockets, elementary TCP and UDP sockets, TCP client/server programming, I/O multiplexing using `select()` and `poll()`, advanced I/O functions, and Linux routing-table and routing-cache operations.

Course Outcome alignment:

- **CO2 – Transport Layer:** Understand transport-layer services and protocols, including UDP and TCP.
- **CO3 – Network Layer:** Understand network-layer functions, protocols, addressing, forwarding, routing and QoS.

The uploaded October 2026 model question paper follows this ESE structure: Part A – answer all 8 questions, 3 marks each; Part B – answer one full 9-mark question from each module, with subdivisions. This question bank follows that pattern. Module 2 questions are mapped to CO2 or CO3 as appropriate.

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2. Part A – Short-answer questions (3 marks each)

Answer each question briefly. Use a labelled diagram, example, comparison or calculation wherever relevant.

2.1 A. Transport-layer services and addressing

1. Define the transport layer. Explain how it provides process-to-process communication.
2. Differentiate between host-to-host communication and process-to-process communication.
3. List any three transport-layer services provided to application processes.
4. Define a socket address. Identify the role of an IP address and a port number.
5. Classify port numbers into well-known, registered and dynamic/private ranges.
6. Distinguish between a server port number and an ephemeral port number.
7. Explain encapsulation and decapsulation at the transport layer.
8. What are multiplexing and demultiplexing? State where each operation takes place.
9. Differentiate between flow control and error control.
10. Define sequence number and acknowledgement number in reliable data transfer.
11. What is a sliding window? State its purpose in flow control.

12. Differentiate between connection-oriented and connectionless services.

2.2 B. Transport protocols: Stop-and-Wait, GBN and SR

13. Explain the operation of the Stop-and-Wait protocol when a data packet is lost.
14. How does Stop-and-Wait handle a lost acknowledgement? Explain the role of the timer.
15. Why are sequence numbers required in Stop-and-Wait? State the number of sequence bits used.
16. Define pipelining in reliable data transfer. State one advantage over Stop-and-Wait.
17. Differentiate between Go-Back-N (GBN) and Selective Repeat (SR) protocols.
18. State the maximum sender-window size for Go-Back-N when the sequence-number field has m bits.
19. State the maximum sender- and receiver-window sizes for Selective Repeat when the sequence-number field has m bits.
20. What is piggybacking? Explain how it improves bidirectional data transfer.

2.3 C. UDP and TCP

21. Define UDP. State any three characteristics of UDP.
22. Draw the UDP header and name its four fields.
23. Calculate the total UDP length when the data field contains 500 bytes.
24. Explain the purpose of the UDP checksum.
25. Differentiate between TCP and UDP with respect to connection setup, reliability and overhead.
26. Define TCP byte numbering. How is the sequence number of a TCP segment determined?
27. What is the purpose of the acknowledgement number in TCP?
28. List any four fields in the TCP header and state their functions.
29. Explain the purpose of the SYN, ACK, FIN and RST control flags.
30. What is a TCP three-way handshake? Name the three segments exchanged.
31. Why does a SYN segment consume one sequence number?
32. Define TCP flow control. What is the role of the advertised receive window?
33. What is silly window syndrome? Name one technique used to avoid it.
34. State the functions of Nagle's algorithm and Clark's solution.
35. Distinguish between TCP flow control and TCP congestion control.
36. What is the purpose of the retransmission timeout (RTO)? State the role of RTT estimation.
37. What is Karn's algorithm? Why is exponential backoff used?
38. Name the major phases of TCP congestion control and briefly state the purpose of slow start.

2.4 D. Network-layer introduction and IPv4

39. State any three functions of the network layer.
40. Differentiate between forwarding and routing.
41. Compare the datagram and virtual-circuit approaches to packet switching.
42. Define network delay. Name the major components of total packet delay.
43. Define throughput and packet loss in a packet-switched network.
44. What is network congestion? Distinguish between open-loop and closed-loop congestion control.
45. List the major components of a router.
46. State the purpose of the IPv4 Time-to-Live (TTL) field.
47. Define MTU. Why is IP fragmentation required?
48. Explain the roles of the Identification, Flags and Fragment Offset fields in IPv4 fragmentation.
49. Differentiate between classful and classless addressing.
50. Define CIDR notation. What information is conveyed by the prefix length?
51. What is subnetting? State one advantage of subnetting.
52. Define address aggregation and longest-prefix matching.
53. Explain the purpose of DHCP. List the four messages exchanged during its basic operation.
54. Define NAT. How does Port Address Translation (PAT) allow multiple private hosts to share a public IP address?
55. State the purpose of ICMPv4. Give two examples of ICMP messages.
56. Explain how ping and traceroute use ICMP messages.
57. What is MPLS? State the purpose of label-based forwarding.

2.5 E. Unicast routing

58. Define unicast routing. What is meant by a least-cost path?
59. Represent a computer network as a weighted graph. Identify the meaning of nodes, edges and link costs.
60. State the Bellman–Ford equation used in distance-vector routing.
61. What is the count-to-infinity problem in distance-vector routing?
62. Explain split horizon and poisoned reverse.
63. What is a link-state database (LSDB)? How is it distributed among routers?
64. State the role of Dijkstra's algorithm in link-state routing.
65. Differentiate between distance-vector and link-state routing.
66. Compare RIP, OSPF and BGP with respect to their routing approach and domain of operation.

67. What is an Autonomous System (AS)? Define an Autonomous System Number (ASN).
68. Differentiate between intradomain and interdomain routing.
69. State the purpose of OSPF areas.
70. What is the role of eBGP and iBGP?

2.6 F. Multicast routing

71. Differentiate between unicast, multicast and broadcast delivery.
72. State two advantages of multicasting over multiple unicasting.
73. What is an IPv4 multicast address? Give its address range.
74. Explain the purpose of IGMP in multicast communication.
75. Differentiate between a source-based tree and a group-shared tree.
76. What is Reverse Path Forwarding (RPF)?
77. State the roles of DVMRP, MOSPF and PIM in intradomain multicast routing.
78. Differentiate between PIM Dense Mode and PIM Sparse Mode.

2.7 G. IPv6, QoS and Linux routing

79. State two reasons for introducing IPv6. Mention the length of an IPv6 address.
80. Name the fields in the IPv6 base header and state the purpose of any three.
81. Differentiate between IPv4 and IPv6 fragmentation.
82. Explain dual stack and tunneling as IPv4-to-IPv6 transition strategies.
83. State the purpose of ICMPv6 and name two Neighbor Discovery messages.
84. Define Quality of Service (QoS). List its four traditional flow characteristics.
85. Differentiate between FIFO queuing, priority queuing and weighted fair queuing.
86. Compare leaky bucket and token bucket traffic-shaping mechanisms.
87. Differentiate between Integrated Services (IntServ) and Differentiated Services (DiffServ).
88. What are the functions of the meter, marker, shaper and dropper in a DiffServ traffic conditioner?
89. What is a Linux routing cache? Distinguish between a cache hit and a cache miss.
90. Define Forwarding Information Base (FIB) and Longest Prefix Matching (LPM).
91. State the purpose of the Linux commands `ip route`, `ip route get` and `ip route add`.
92. What is a socket? Name the basic system calls used in a TCP server and a TCP client.
93. What is I/O multiplexing? State the purpose of the `select()` and `poll()` functions.

3. Part B – Descriptive / analytical questions (9 marks each)

Each question is designed as a full 9-mark option. Subdivision marks are indicated. Draw neat, labelled diagrams wherever required. For numerical questions, show all intermediate steps.

3.1 A. Transport-layer services and reliable data transfer

1. Explain the services provided by the transport layer. Describe process-to-process delivery, port addressing, multiplexing, demultiplexing, encapsulation and decapsulation with a suitable diagram. (5+4 marks)

2. Explain the Stop-and-Wait protocol using sender and receiver timelines. Discuss how the protocol handles a lost data packet, a lost acknowledgement and duplicate frames. (6+3 marks)

3. Explain the Go-Back-N protocol with a neat diagram. Describe the sender window, receiver window, sequence-number operation, cumulative acknowledgements and retransmission after a timeout. (6+3 marks)

4. Explain the Selective Repeat protocol. Describe its sender and receiver windows, individual acknowledgements, buffering of out-of-order packets and retransmission mechanism. (6+3 marks)

5. Compare Stop-and-Wait, Go-Back-N and Selective Repeat protocols in terms of pipelining, window size, receiver behaviour, acknowledgement and retransmission. (5+4 marks)

6. A reliable data-transfer protocol uses $m = 3$ bits for sequence numbers. Determine the sequence-number range and the maximum sender-window size for GBN and SR. Justify the window-size restrictions. (5+4 marks)

7. Explain flow control and error control at the transport layer. Discuss sequence numbers, acknowledgements, timers, retransmissions and sliding-window operation. (5+4 marks)

3.2 B. UDP and TCP

8. Draw and explain the UDP header. Describe the function of the source port, destination port, length and checksum fields. Calculate the UDP length for a 1200-byte application message. (6+3 marks)

9. Explain the services and characteristics of UDP. Discuss why UDP is suitable for applications requiring low overhead and delay-sensitive communication. Give suitable application examples. (6+3 marks)

10. Explain the TCP segment format with a neat diagram. Describe the purpose of the important header fields and control flags. (6+3 marks)

11. Explain TCP byte numbering and acknowledgement numbering with a suitable example. Show how sequence and acknowledgement numbers are assigned when data is transmitted in both directions. (5+4 marks)

12. Explain TCP connection establishment using the three-way handshake. Draw the sequence diagram and justify why a SYN segment consumes one sequence number. (6+3 marks)

13. Describe TCP data transfer and connection termination. Explain the exchange of FIN and ACK segments and the purpose of the TIME-WAIT state. (6+3 marks)

14. Explain TCP flow control using the sender and receiver windows. Discuss advertised window, window opening and closing, window shrinking and zero-window operation. (6+3 marks)

15. What is silly window syndrome? Explain how Nagle's algorithm and Clark's solution help prevent it. (3+6 marks)

16. Explain TCP error control. Discuss checksum, acknowledgements, retransmission, duplicate acknowledgements and handling of out-of-order segments. (5+4 marks)

17. Explain TCP congestion control. Describe slow start, congestion avoidance and the AIMD principle with suitable congestion-window behaviour. (6+3 marks)

18. Explain the purpose of TCP timers. Discuss the retransmission timer, RTT and RTO estimation, Karn's algorithm, exponential backoff, persistence timer, keepalive timer and TIME-WAIT timer. (6+3 marks)

19. A TCP sender transmits a segment with sequence number 1200 and 500 bytes of data. Find the sequence number of the next byte and the acknowledgement number expected from the receiver if all bytes are received correctly. Explain your answer. (5+4 marks)

3.3 C. Network-layer introduction, IPv4 and forwarding

20. Explain the services of the network layer. Discuss packetizing, routing, forwarding, error control, congestion control and Quality of Service. (6+3 marks)

21. Compare datagram and virtual-circuit packet-switching approaches. Explain the setup, data-transfer and teardown phases of a virtual circuit. (5+4 marks)

22. Explain the components of total delay in a packet-switched network. Discuss processing, queuing, transmission and propagation delays, throughput and packet loss. (6+3 marks)

23. Explain the structure and functions of a router with a neat diagram. Describe input ports, switching fabric, output ports and routing processor. (6+3 marks)

24. Draw the IPv4 datagram format and explain the function of its major header fields. (6+3 marks)

25. An IP datagram of total length 4000 bytes, including a 20-byte header, must pass through a link with an MTU of 1500 bytes. Find the number of fragments. For each fragment, calculate total length, fragment offset and MF flag. (9 marks)

26. Explain classless addressing and CIDR notation. Describe how to obtain the network address, block size, address range and broadcast address from a CIDR address. (5+4 marks)

27. An organization is assigned the address 192.168.10.75/26. Find the subnet mask, network address, broadcast address, usable host range and number of usable host addresses. (9 marks)

28. Explain subnetting and address aggregation with suitable examples. Discuss how CIDR and route summarization reduce forwarding-table size. (5+4 marks)

29. Explain the IP forwarding process using a forwarding table. Describe prefix matching and the longest-prefix matching rule with a suitable example. (6+3 marks)

30. Explain the operation of DHCP. Describe the DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACK messages and the information supplied to a client. (6+3 marks)

31. Explain Network Address Translation (NAT). Describe NAT using one public address, a pool of public addresses and Port Address Translation (PAT), with suitable translation-table examples. (6+3 marks)

32. Explain ICMPv4 message categories and format. Describe how ping and traceroute use Echo Request, Echo Reply and Time Exceeded messages. (5+4 marks)

33. Explain the working of MPLS. Describe label-based forwarding, the MPLS header fields and the use of a label stack. (6+3 marks)

3.4 D. Unicast routing

34. Explain the distance-vector routing algorithm using the Bellman–Ford equation. Describe how routers exchange distance vectors and update their routing tables. (6+3 marks)

35. Explain the count-to-infinity problem in distance-vector routing. Discuss two-node loops, split horizon and poisoned reverse as solutions. (5+4 marks)

36. Explain link-state routing. Describe neighbor discovery, link-state packet generation, flooding, construction of the link-state database and shortest-path computation. (6+3 marks)

37. Apply Dijkstra's algorithm to a weighted network to find the least-cost paths from a specified source router to all other routers. Construct the routing table showing destination, cost and next hop. (9 marks)

38. Compare distance-vector and link-state routing with respect to information exchanged, algorithm, convergence, resource requirements and examples. (5+4 marks)

39. Explain RIP as an intradomain distance-vector routing protocol. Discuss its metric, request/response messages, update process and timers. (6+3 marks)

40. Explain OSPF as an intradomain link-state routing protocol. Describe areas, link-state advertisements, neighbor discovery and the role of Dijkstra's algorithm. (6+3 marks)

41. Explain BGP4 as an interdomain path-vector routing protocol. Differentiate eBGP and iBGP and discuss path attributes and route selection. (6+3 marks)

42. Compare RIP, OSPF and BGP4 with respect to algorithm, scope, metric/path information and typical use. (5+4 marks)

3.5 E. Multicast routing

43. Explain multicast communication and compare it with unicast and broadcast. Discuss the advantages of multicast over multiple unicasts. (5+4 marks)

44. Explain IPv4 multicast addressing and the role of IGMP in collecting group-membership information. Describe the basic join and leave operations. (6+3 marks)

45. Explain source-based multicast trees and group-shared multicast trees. Describe the role of a core or rendezvous point in a group-shared tree. (5+4 marks)

46. Explain Reverse Path Forwarding (RPF), Reverse Path Broadcasting (RPB) and Reverse Path Multicasting (RPM). Show how they help construct multicast distribution trees. (6+3 marks)

47. Explain DVMRP and MOSPF as intradomain multicast routing protocols. (5+4 marks)

48. Explain Protocol Independent Multicast. Compare PIM Dense Mode and PIM Sparse Mode with respect to forwarding, pruning, group membership and tree construction. (4+5 marks)

49. Discuss interdomain multicast routing. Explain the roles of MBGP and MSDP as described in the study material. (6+3 marks)

3.6 F. IPv6 and Quality of Service

50. Explain the IPv6 base-header format with a neat diagram. Describe the Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Source Address and Destination Address fields. (6+3 marks)

51. Compare IPv4 and IPv6 with respect to address size, header format, fragmentation, options and supporting protocols. (6+3 marks)

52. Explain IPv6 address representation. Describe colon-hexadecimal notation, leading-zero abbreviation and zero compression with examples. (5+4 marks)

53. Explain the three IPv6 transition strategies: dual stack, tunneling and header translation. State the situation in which each strategy is used. (6+3 marks)

54. Explain the purpose of ICMPv6. Describe Neighbor Discovery and the roles of Router Solicitation, Router Advertisement, Neighbor Solicitation and Neighbor Advertisement. (6+3 marks)

55. Define QoS. Explain reliability, delay, jitter and bandwidth as the four traditional flow characteristics. Relate these characteristics to different applications. (5+4 marks)

56. Compare FIFO queuing, priority queuing and weighted fair queuing. Discuss the starvation problem associated with priority queuing. (6+3 marks)

57. Explain traffic shaping and policing. Describe the operation of leaky bucket and token bucket algorithms with suitable examples. (5+4 marks)

58. Explain Integrated Services (IntServ). Discuss flow specification, Guaranteed Service, Controlled-Load Service and the role of RSVP Path and Resv messages. (6+3 marks)

59. Explain Differentiated Services (DiffServ). Describe per-class treatment, DS field, default/EF/AF per-hop behaviours and traffic-conditioning components. (6+3 marks)

60. Compare IntServ and DiffServ with respect to service model, state maintained by routers, scalability and resource reservation. (5+4 marks)

3.7 G. Linux routing and socket programming (syllabus hands-on)

61. Explain the Linux routing-table lookup path using the routing cache and Forwarding Information Base (FIB). Differentiate between a cache hit and a cache miss. (5+4 marks)

62. Describe the purpose of `rt_hash_bucket`, `rtable`, `dst_entry` and the FIB structures used in the supplied Linux routing-cache notes. (6+3 marks)

63. Explain how a route is added to the Linux kernel using the `ip route add` command. Trace the request through RT Netlink to the route-insertion handler and FIB. (6+3 marks)

64. Write the Linux commands to display IP addresses, network interfaces and the routing table; add a route through a gateway; query the route to a destination; and delete a route. Explain the purpose of each command. (6+3 marks)

65. Explain the basic socket-programming model for TCP client/server communication. Describe the roles of `socket()`, `bind()`, `listen()`, `accept()`, `connect()`, `send()/write()` and `recv()/read()`. (6+3 marks)

66. Explain elementary UDP socket communication. Compare the operation of a UDP client/server with a TCP client/server. (5+4 marks)

67. Explain I/O multiplexing. Describe how `select()` and `poll()` allow a server to monitor multiple descriptors. (6+3 marks)

4. Model-question-paper style practice sets

The following practice sets reproduce the Module 2 Part B pattern in the uploaded model paper: two alternatives, each carrying 9 marks. The questions are distributed across Transport Layer and Network Layer topics.

4.1 Practice Set 1 – Module 2

Q1 (a) Explain TCP connection establishment using the three-way handshake with a neat sequence diagram. Justify why a SYN segment consumes one sequence number. (6)
(b) Select TCP or UDP for an online fund-transfer application, live video-conferencing and a DNS query. Justify each choice. (3)

OR

Q2 (a) Explain the IPv4 fragmentation process and the roles of Identification, Flags and Fragment Offset. (6)
(b) Define MTU and state why fragmentation may be required. (3)

4.2 Practice Set 2 – Module 2

Q1 (a) Explain TCP flow control using sender and receiver windows. Discuss the advertised window and zero-window condition. (6)
(b) Differentiate between flow control and congestion control. (3)

OR

Q2 (a) Explain distance-vector routing using the Bellman–Ford equation. Discuss the count-to-infinity problem. (6)
(b) Explain split horizon and poisoned reverse. (3)

4.3 Practice Set 3 – Module 2

Q1 (a) All routers have the same link-state database for a weighted network. Apply Dijkstra's algorithm from a given source router and construct its routing table showing destination, cost and next hop. (6)
(b) State two differences between distance-vector and link-state routing. (3)

OR

Q2 (a) An IPv4 datagram of total length 4000 bytes, including a 20-byte header, must pass through a link with an MTU of 1500 bytes. Calculate the number of fragments and the total length, fragment offset and MF flag for each fragment. (6)
(b) Explain the purpose of the IPv4 Identification field. (3)

4.4 Practice Set 4 – Module 2

Q1 (a) Explain the IPv6 base-header format and the purpose of its important fields. (6)
(b) Differentiate between dual stack and tunneling. (3)

OR

Q2 (a) Explain QoS and its four flow characteristics. Describe FIFO, priority and weighted fair queuing. (6)
(b) Differentiate between leaky bucket and token bucket. (3)

4.5 Practice Set 5 – Module 2

Q1 (a) Explain multicast routing using RPF, RPB and RPM. Describe how multicast distribution trees are constructed. (6)
(b) Explain the role of IGMP in multicast group management. (3)

OR

Q2 (a) Explain how a new route is added to the Linux routing table using the `ip route add` command. Trace the RT Netlink and FIB operations described in the supplied hands-on notes. (6)
(b) Write the commands to display the routing table and query the route to a destination. (3)

5. Coverage and CO mapping

Syllabus area	Topics represented in the question bank	CO
Transport-layer services and protocols	Services, port addressing, sockets, multiplexing, flow/error control, Stop-and-Wait, GBN, SR, piggybacking	CO2
UDP and TCP	UDP header/checksum; TCP segment, connection setup/termination, sequence and acknowledgement numbers, flow/error/congestion control and timers	CO2
Network-layer introduction and protocols	Packet switching, delay, throughput, router, IPv4, fragmentation, CIDR, DHCP, NAT, ICMPv4 and MPLS	CO3
Unicast routing	Distance vector, link state, Dijkstra, RIP, OSPF, BGP and autonomous systems	CO3
Multicast routing	Multicast addressing, IGMP, RPF/RPB/RPM, DVMRP, MOSPF, PIM, MBGP and MSDP	CO3
Next-generation IP and QoS	IPv6 header/addressing/transition, ICMPv6, QoS, queuing, shaping, IntServ, RSVP and DiffServ	CO3
Hands-on components	TCP/UDP sockets, client/server, select/poll, Linux routing table, routing cache, FIB and `ip` commands	CO2/CO3